

A classification of well-behaved graph clustering schemes

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1 Introduction

general stuff about clustering and why it is an interesting problem

some talk about why optimization-based methods have poor theoretical guarantees

some talk about why an axiomatic approach gives algorithms that run in polynomial time, unlike the generally NP-hard optimization problems

soft heuristic statement of what our axioms/properties are and why they might be desirable

statement of our results

2 Definitions of our terms

Definition 1. The category of simple graphs $\mathcal{G}$ has as its objects simple graphs, which we consider to be a finite set V equipped with a reflexive and symmetric binary relation E – that is, our simple graphs are finite, have no double edges, and have loops at every vertex¹. A morphism from $H = (V, E)$ to $G = (V', E')$ is an injective set function $f : V \rightarrow V'$ such that whenever uEv , we also have $f(u)E'f(v)$. That is, a morphism from H to G is a way of realizing H as a subgraph of G .

Definition 2. The category of partitioned sets $\mathcal{P}$ has as its objects sets V with an equivalence relation $\sim$. A morphism in $\mathcal{P}$ from $(V, \sim)$ to $(W, \approx)$ is a set function $f : V \rightarrow W$ such that whenever $v \sim w$, $f(v) \approx f(w)$. So in the case where $V = W$, $f = \text{Id}$ is a morphism if $\approx$ is a coarsening of $\sim$.

Definition 3. A clustering scheme is a function² $\mathfrak{C}$ that sends simple graphs to partitioned sets, with the same underlying set. We say that it is *functorial* if it is a functor from $\mathcal{G}$ to $\mathcal{P}$.

Definition 4. A clustering scheme $\mathfrak{C}$ is *excisive* if, for all graphs $G = (V, E)$, it holds for every equivalence class A of $\mathfrak{C}(G)$ that $\mathfrak{C}(G|_A) = (A, A^2)$. That is, if we take a graph, cluster it, pick out one of the parts, and cluster that part alone, that part will be clustered into a single part.

¹The final property here may seem a bit unusual, but it simplifies our proofs in some places, without really affecting much.

²Technically, of course, neither $\mathcal{G}$ nor $\mathcal{P}$ are small categories, so the word “function” should be read as “class function” here, not in the usual sense of the word. We sweep this issue under the rug here, because it will never affect anything.

- 3 Representability
- 4 Proof of equivalence of representability and excisiveness
- 5 Representability for hierarchical clustering